

Combining Augmented and Virtual Reality Experiences for Immersive Fire Drills

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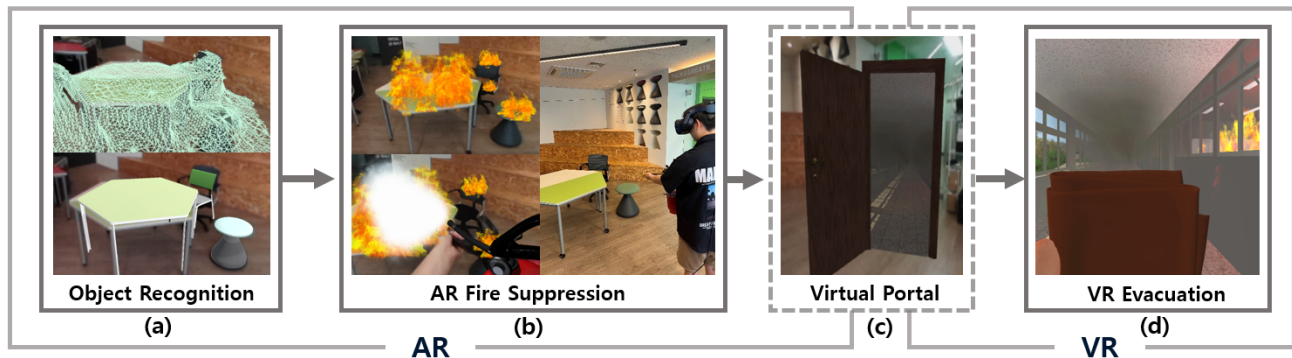


Figure 1: A Fire Drill Scenario Combining AR and VR Experiences

ABSTRACT

We propose an application that combines the benefits of augmented reality (AR) and virtual reality (VR) experiences for realistic and effective fire response training. Our analysis of recent literature reveals that there are different advantages to AR versus VR applications, and combining these experiences can generate new compound benefits in fire drills. In this study, a high-level training environment is automatically generated using three-dimensional object recognition technology, and the physical training tools are linked to the VR environment to improve immersion and training effectiveness. The combined advanced AR and VR experience demonstrates the potential use of this method in various industries.

CCS CONCEPTS

• Human-centered computing → augmented / virtual reality.

KEYWORDS

augmented reality, virtual reality, fire drills

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1 INTRODUCTION

Recently, augmented reality (AR) and virtual reality (VR) technologies have attracted much attention due to the proliferation of non-face-to-face technologies. These techniques can be applied to a wide range of fields, such as the training of emergency procedures. Realistic fire-drill practices are nearly impossible to conduct in daily operational environments, as the risk factors and training costs are prohibitively high. However, these problems can be mitigated by using AR and VR techniques. Contemporary AR/VR fire-drill methods have been studied with demonstrable effectiveness [Chen et al. 2021]. However, the advantages of AR versus VR for realistic training differ. Tarkkanen described the benefits of interacting with users using independent AR and VR applications when teaching fire safety techniques [Tarkkanen et al. 2020]. Additionally, Hitchcock showed the potential of this combination by proposing interesting applications that integrated the unique and desirable features of AR and VR [Hitchcock and Sung 2018]. However, previous studies have created an artificial environment and remain at the level of simple experience. In this paper, we propose a fire-drill application that combines AR and VR to achieve safe and valid upskilling while creating an immersive and effective training environment. First, a realistic fire is simulated based on the type, size, and shape of real objects in the environment. Second, a training-purposed real fire extinguisher prototype is synchronized with the virtual environment. Several studies have produced highly effective training effects using such tangible user interfaces [Nathoo et al. 2020]. Finally, we introduce a smart method of switching between the AR

and VR environments using a single head-mounted display (HMD) for AR fire suppression and VR evacuation training.

2 OUR APPROACH

2.1 Overview

Figure 1 shows the pipeline of our innovative fire-drill application. The user wears a VIVE PRO HMD, which quickly conducts a three-dimensional scan of the environment to identify the constituent objects in the scene based on their type, size, shape, and location (Fig. 1-a). Virtual fire effects are then generated based on this data, and the user can suppress the fire using the tangible fire-extinguisher mock-up (Fig. 1-b). After AR suppression training, the trainee switches to VR evacuation training through a software portal (Fig. 1-c). During evacuation training, trainees must lower their head, cover their mouth with a virtual handkerchief, and escape by walking in place along the virtual escape route (Fig. 1-d).

Trainees used a Desktop PC (Ryzen7 5800x, GTX 1080Ti, 32GB RAM), VIVE PRO HMD, and a tangible fire extinguisher prototype. The SRWorks v.0.9.7.1 software development kit (SDK) provided by VIVE, was used to support AR and VR via the HMD. The software environment consisted of the Unity 3D engine (2019.4.29.f1).

2.2 Creating the AR Fire Training Environment

For AR suppression training, it is crucial to identify objects in the environment that may become ignited during a fire. In this study, two object recognition methods were integrated. The first used the artificial intelligence (AI) vision module supported by the SRWorks SDK. This module can recognize up to 20 object types, including chairs and tables. The second method used the mobilenet-V2 deep learning model to further classify the recognized objects. This integrated approach resulted in a realistically rendered virtual fire scene based on the shape and sizes of real objects.

2.3 Interface for AR Fire Training

The prototype was used to build a tangible water fire extinguisher (4L) product. Various sensors were used to synchronize the tangible user interface with the AR training environment (Fig. 2-a). The direction and position of the fire extinguisher body and hose were synchronized with the virtual environment, and a virtual powder was sprayed when the prototype extinguisher handle was gripped. An Arduino (UNO WiFi R2) was used to control these sensors wired or wirelessly. AR fire training using this prototype proved its effectiveness through system usability test.

2.4 VR Fire Evacuation Training

To proceed from AR to VR training, the two environments were switched using a virtual portal (Fig. 1-c). An essential point during VR evacuation training is how to move and behave. Notably, a user can move naturally while wearing an Omni Tracker on both ankles and by walking in place (Fig. 2-b). And, the desired behavior included covering one's nose and mouth with a virtual handkerchief while lowering the head and upper body to avoid the inhalation of toxic gases. The HMD's six-degree-of-freedom function tracks the user's upper body height and synchronizes another VIVE Tracker

with a virtual handkerchief to ensure that the nose and mouth are covered.

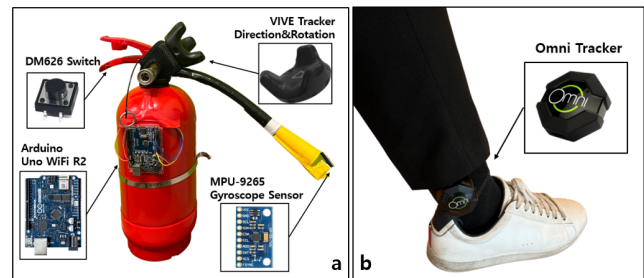


Figure 2: (a) fire-extinguisher prototype, (b) locomotion with omni tracker

3 CONCLUSION AND FUTURE WORK

This paper presented a training application that combines the strengths of AR and VR for realistic fire suppression and evacuation training. Specifically, the fire-training environment was constructed by accurately detecting the 3D information of real objects to improve the immersive experience of the training. By applying the actually used fire training tool to the tangible interface, the training effect was increased and this was verified. Finally, a method for freely switching between AR and VR experiences in one device was presented. The proposed system still has limitations that can only be used on a few devices. In the future, we plan to develop human-like AI agents that can respond to various training scenarios alongside the trainee, further reducing risk and reducing the cost of human participants.

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